CHAPTER

28

Minimal Intervention Dentistry

In the recent past, scientific developments in cariology, dental materials and diagnostic armamentarium have revolutionized operator's approach in managing caries. The previous surgical approach requiring removal of diseased portion of tooth along with extension of cavities to the areas prone to caries is no longer used. The '*extension for prevention*', which was commonly taught has been changed to '*prevention for extension*'. The tooth weakened by extensive surgical approach leads to frequent retreatment and replacement, subsequently resulting in more complex treatment modalities and even extraction. The emerging technologies have paved the way for preventive aspects vis-à-vis remineralization of the existing lesions. Though this concept might not be fully practical, however, the early trials are promising. Alternatively, the treatment of the existing lesion should include minimum involvement of the healthy tooth structure.

The terms '*minimal intervention dentistry*' and '*minimal invasive dentistry*' are used interchangeably, although they describe different clinical concepts. Minimal intervention dentistry implies patient care that deals with the cause of dental disease and not just the symptom. Minimal intervention is based entirely on prevention and control of oral disease; whereas, minimal invasive dentistry embraces conservative restorative procedures (Flowchart 28.1). The excavation of caries is performed with the objective of preserving not only sound tooth tissues but also that tissue, which has potential to remineralize. The use of adhesive biomaterial is preferred. Minimal invasive dentistry is merely a component of minimal intervention treatment (care) plan. The terms, '*ultraconservative*' and '*microdentistry*' are synonymous with minimal invasive dentistry, that implies use of magnification aids and other devices, such as, chemo-mechanical, air abrasive, sono-abrasion and laser systems for management of caries.

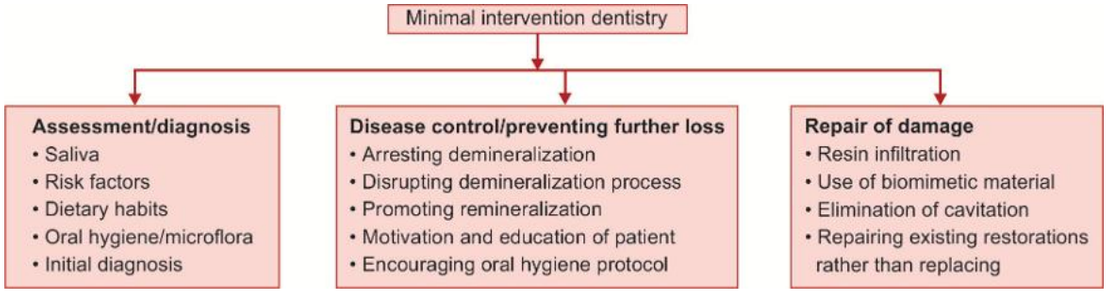
Core Principles

The basis of minimal invasive dentistry is to deviate the clinical focus from restoring the teeth damaged by caries to the prevention of disease in its early stages.

The core principles of minimal invasive dentistry are:

i. *Recognizing risk factors:* To assess the potential caries risk factors, may undertake some diagnostic tests.
ii. *Minimizing risk factors:* The risk factors are minimized by reducing cariogenic diet, altering fluid balance and managing pH of the oral environment.
iii. *Regeneration/remineralization:* The incipient lesions are arrested by remineralizing agents, such as fluorides, casein phosphopeptide-amorphous calcium phosphate (CPP-ACP).

Flowchart 28.1: Minimal intervention dentistry



iv. *Repair:* In case of cavitation, minimum tooth structure is to be sacrificed (use of conservative means). Bioactive materials are preferred to restore teeth where ample amount of dentin is lost. The old restorations are repaired, rather replacing with newer restorations.

The principles of minimal intervention in managing dental caries as adopted by FDI are:

- Patient education and motivation
- Modification of oral micro-organisms
- Remineralization of early caries lesions
- Minimal tooth preparation for cavitated lesions
- Repair of defective restorations

The minimal invasive approach in managing dental caries incorporates detecting, diagnosing, intercepting and treating dental caries on the microscopic level. This approach of treating dental caries includes non-surgical modalities or minimum surgical intervention. However, the basic concept of dental caries being infectious disease, is followed.

The focus of minimal invasive dentistry is on maximum conservation of demineralized, non-cavitated lesions of enamel and dentin. The development of adhesive dentistry and scientific knowledge in understanding the progress of caries coupled with recent diagnostic aids have enabled the operator to carry out precise tooth preparation than just mechanical removal of the tooth tissue.

COMPONENTS OF MINIMAL INVASIVE APPROACH

The components of minimal invasive approach include the following:

I. Early diagnosis of caries
II. Caries risk assessment (young and old)
III. Strategies to arrest existing carious lesions
IV. Remineralization of the existing lesions
V. Restoring cavitated lesions: Minimal invasive approach
VI. Caries inhibition by resin infiltration
VII. Repair rather than replacement of defective restorations

I. EARLY DIAGNOSIS OF CARIES

The early diagnosis involves detection of carious lesions at initial stages and also determine the caries activity. Caries activity must be evaluated by monitoring the lesion over time. Various caries activity tests are available. Indices for progress of caries activity are also used; however, improved versions are required to analyze the progress of the caries. (For details, refer author's book 'Dental caries': Vimal K Sikri.)

Various diagnostic aids are being utilized to detect caries. Some are preferred for detecting occlusal caries, while others detect proximal or smooth surface lesions in a better way. The emerging technologies in diagnostic armamentarium include electrical conductance methods, quantitative laser fluorescence, tuned-aperture computed tomography and optical coherence tomography (For details refer Chapter 9). Radiographs, especially bitewing, are also helpful in analysing progress of caries; however, not useful in detecting early caries.

There is still a need for research to increase the accuracy of diagnostic methods. The on-the-spot assessment of caries activity is required to effectively perform minimal invasive procedures.

II. CARIES RISK ASSESSMENT (YOUNG AND ADULTS)

Risk assessment helps professionals supervise individual's oral health. It is a process that attempts to identify those individuals who are at greater risk of caries attack. These features are helpful in early treatment of caries and also the potential of reversal phenomenon.

The patients are categorized as 'low risk', 'moderate risk', and 'high risk', depending upon the risk of development of caries.

Low risk

- Good oral hygiene
- No new caries in the last couple of years
- No susceptible pits and fissures
- Visiting dental clinic regularly

Moderate risk

- Oral hygiene fair
- One caries lesion in last year
- Susceptible pits and fissures
- Irregular visit to dental clinic
- Undergoing orthodontic treatment

High risk

- Poor oral hygiene
- Two or more than two carious lesions in last year
- Susceptible (deep) pits and fissures
- Inadequate salivary flow
- Rarely visit dental clinic
- Inappropriate and frequent bottle feeding

The procedure involves examining risk factors that may negatively impact individuals oral health, and protective factors that promote oral health. Using risk assessment, the operator can make specific preventive and treatment recommendations to improve overall oral health. This also contributes to efficient management of case by eliminating unnecessary interventions.

Risk Factors in Caries Initiation

The risk factors for the initiation of caries can be physical, biological, environmental, and behavioral. Other factors include number of cariogenic bacteria, compromised salivary flow, poor oral hygiene and frequent consumption of cariogenic food.

The important factors are as follows:

a. *Host (Tooth) Factors*

The host factors may influence the initiation of caries in many ways. First, some areas of the tooth, such as pit and fissures, tooth irregularities, etc. are prone to biofilm accumulation. Second, quality and quantity of saliva, buffering capacity, flow, antimicrobial activity and also clearance from the oral cavity.

b. *Biofilm (Bacterial Stagnation)*

Many biofilm micro-organisms have acid producing and acid tolerating properties, including species from the genera *Veillonella*, *Lactobacillus*, *Bifidobacterium*, *streptococci*, *Actinomyces* spp., etc.

Lower pH in biofilm may not result in progressive net mineral loss. The prolonged acidification of biofilm may result in mineral loss and subsequent caries. The biofilm environment, which favors more acid producing micro-organisms, facilitates initiation of caries.

c. *Dietary Factors*

The dietary sugars are considered as the main causative agents in the initiation of caries. Diet high in fermentable carbohydrates favor a low biofilm pH, while diet high in proteins and fats favor a neutral biofilm pH. High-protein diets increase the urea concentration of saliva, which can be converted by ureolytic bacteria to ammonia; this raises the biofilm pH and is associated with decreased caries risk.

Other dietary factors, such as type of carbohydrate, retentiveness and presence of protective factors in food (calcium, phosphate and fluoride), are important. Complex carbohydrates (starches) are considered less cariogenic than the simple sugars (sucrose, glucose, and fructose) with sucrose being the most cariogenic, because of its ability to produce extracellular glucans. Amino acids and proteins have the potential of being 'anti-caries', and have significant caries-inhibitory and cariostatic effects.

i. ***Effect of minerals:*** Trace elements (pre-eruptively/post-eruptively), when incorporated in enamel, can affect remineralization process and influence the metabolism of cariogenic micro-organism. The elements, which promote caries are Se, Mg and Cd; while, Mo, V, and Sr, are mildly cariostatic.

ii. ***Effect of vitamins:*** Vitamins A, C, and D are essential for proper calcification of tooth structure. The vitamin–caries relationship is mediated primarily through acidogenic micro-organisms. Vitamin B is necessary growth factors for most of the oral acidogenic flora. Thiamine and nicotinic acid may promote acid production by microbial action.

d. *Micro-organisms*

High proportion of mutans streptococci is essential for initiation of caries. *Streptococcus mutans* produces high levels of acid from sugars, is acid tolerant and firmly adheres to tooth surface. Lactobacilli and other gram-positive bacteria, commonly isolated from the oral cavity, are high acidogenic, but they have low affinity to the tooth surface. (*Lactobacilli and gram-positive bacteria are more related to advancing caries rather than initiating caries.*) The acid produced during the metabolic activities cause demineralization, which increases porosity in the enamel and decreases translucency of the surface color.

Caries risk factors: General features

Physical
- Variations in tooth enamel, deep pits and fissures, anatomically susceptible areas
- Gastric reflux
- High mutans streptococci count
- Previous caries experience
- History of baby bottle tooth decay

Behavioral
- Feeding bottle used at night for sleep; or 'at will' while awake
- Taking frequent snacks
- Inadequate oral hygiene
- Eating disorders, including self-induced vomiting (bulimia)

Socio-environmental
- Inadequate fluoride
- Poor family oral health
- High parental levels of mutans streptococci

Disease or treatment related
- Cariogenic carbohydrate diet
- Disturbed salivary flow due to medication/ irradiation
- Orthodontic appliances

III. STRATEGIES TO ARREST EXISTING CARIOUS LESIONS

The strategy involves use of certain chemical agents, which can decrease the rate of progress of the carious lesion. Mostly fluorides are utilized in arresting the existing lesions; help remineralization of the affected part of the dentin (Fig. 28.1) and arrest the progress of the lesion.

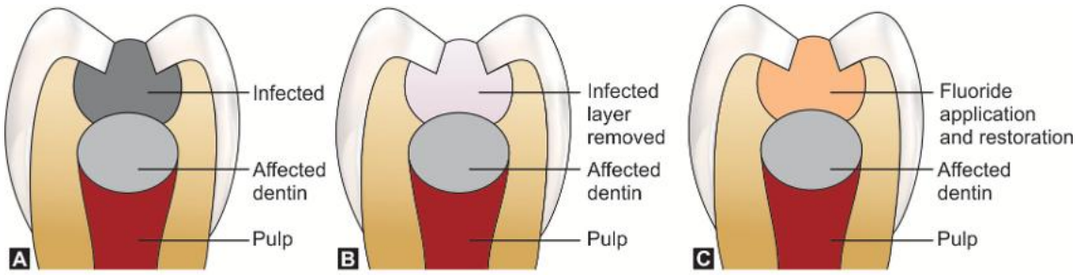


Fig. 28.1: Arresting caries progression: **(A)** Active caries; **(B)** Removing infected caries; **(C)** Fluoride application over the affected dentin to arrest caries progress

Silver diamine fluoride [$Ag(NH_3)_2F$], a combination of silver nitrate and sodium fluoride, when applied to carious dentin inhibits the progress of lesion. 38% silver diamine fluoride, used in cavitated carious lesions, has shown promising results. Twice yearly application of this concentration might be able to arrest the carious process. This non-invasive treatment has the advantage that trained dental auxiliary personnel can also apply the solution.

Ozone gas (an antimicrobial agent) also reduces the number of micro-organisms on tooth surfaces. It is naturally produced in the presence of light or by different industrial processes. Ozone has the potential to kill cariogenic bacteria and subsequently leads to arrest of carious lesion. An ozone generating system, 'KaVo Healozone' (delivers ozone gas at a concentration of approximately 2100 ppm), has been developed. The application procedure normally takes between 20 and 120 seconds per tooth surface.

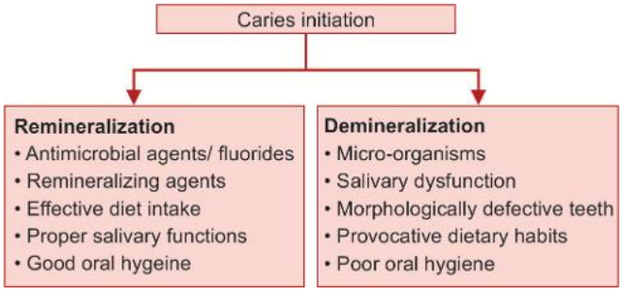
IV. REMINERALIZATION OF THE EXISTING LESIONS

The possibility of arresting and even reversing the mineral loss associated with initiation of caries has been established. Depending upon the oral environment, the tooth alternatively loses and gains calcium and phosphate ions. This cyclic loss and gain forms the basis of demineralization and remineralization (Flowchart 28.2). The pH of the environment surrounding the teeth is an important factor for this process.

Taking advantage of the tooth's ability to remineralize, the balance can be tipped in favor of remineralization by altering the oral environment. The phenomenon of alteration includes:

- Decreasing the frequency of intake of refined carbohydrates
- Following optimum plaque control measures
- Maintaining salivary flow
- Motivating and educating the patient

Flowchart 28.2: Basis of caries initiation



Ideal Requirements of a Remineralizing Agent

- Diffuse into subsurface, deliver appropriate amount of calcium and phosphorus (should not diffuse excess of calcium)
- Should not encourage calculus formation
- Should be effective in xerostomic patients
- Reinforce the remineralizing component of saliva
- Should be effective in acidic pH
- Should follow simple/easy technique

A. Remineralization of Enamel Lesions

The materials used for enamel remineralization are as follows.

a. *Fluorides*

The fluorides, both as systemic and topical application, increase caries resistance. The action of fluorides is summarized as: Increased enamel resistance, remineralization of incipient caries, controlling micro-organisms and increase in rate of maturation.

Fluorides present in oral environment alter the continuously occurring dissolution at the tooth–oral fluids interface.

Topical fluorides encourage:

- Remineralization at the crystal suface by making them acid resistant.
- Inhibition of bacterial enzymes.

Fluorides facilitate formation of fluorapatite, which is deposited at the surface layer, while hydroxyapatite is dissolved from the subsurface. Fluorides have their limitation in providing calcium and phosphate ions from oral biofilm to the demineralized enamel. Also the effect of fluorides is limited in pit and fissure

caries. Higher concentration of fluoride leads to deposition of calcium fluoride on the surface, releasing fluoride slowly. The rate of fluoride release is enhanced at lower pH level. Extra release of fluoride is not considered conducive for overall health of enamel. These limitations prompted researchers to look for alternative remineralizing agents.

b. *Casein Derivatives: Casein Phosphopeptide-Amorphous Calcium Phosphate (Recaldent)*

Casein is the predominant phosphoprotein in bovine milk and accounts for almost 80% of its total protein. The calcium phosphate is present as an alkaline amorphous phase along with calcium phosphate, referred to as casein phosphopeptide-amorphous calcium phosphate (CPP-ACP). They have the ability to stabilize high concentrations of calcium and phosphate in metastable solutions. CPP-ACP is a useful cariostatic agent for the control of dental caries as it contains calcium and phosphate in a bioavailable form. CPP-ACP can be used as an:

- Adjunct preventive therapy to reduce caries in high risk patients
- Reduce dental erosion in patients with gastric reflux
- Remineralize enamel involving initial white spot lesions
- Minimize orthodontic decalcification
- Potent desensitizing agent

CPP-ACP binds readily to the surface of tooth; the bound CPP-ACP buffers the free calcium and phosphate ions and increases level of calcium phosphate in plaque. This saturation inhibits enamel demineralization and enhances remineralization. CPP-ACP is pH responsive. Increase in pH increases the level of bound ACP; stabilizes free calcium and phosphate in plaque; subsequently, inhibiting acidic activities in plaque. CPP-ACP influences the behavior of plaque through:

- Binding to adhesin molecule of *S. mutans* and impairing their incorporation into plaque.
- Elevates plaque-calcium ion level; inhibiting acidic activities of plaque (plaque fermentation).
- Suppress overgrowth of aciduric species, where fermentable carbohydrates are in excess (provide protein and phosphate buffering of plaque pH).

Casein derivatives added in toothpastes, dentifrices and gum/lozenges have shown promising results, inhibiting caries and enhancing remineralization. CPP-ACP has also been used with fluoride for better results (CPP-ACFP: amorphous calcium fluoride phosphate).

c. *Tricalcium Phosphate (TCP)*

Tricalcium phosphate, precisely β-tricalcium phosphate (βTCP), substantially enhances levels of calcium in plaque and saliva. The commercial preparations are Cerasorb, Bio-resorb, etc. β-TCP is less soluble than α-TCP; less likely to provide bio-available calcium. The common problem using TCP is the formation of calcium–phosphate complexes (with fluorides, formation of calcium fluoride), which would inhibit remineralization by lowering the levels of bioavailable calcium and fluoride. TCP can be combined with titanium dioxide (or any metal oxide) to limit the interaction between calcium and phosphate, making the material more stable in oral conditions. When TCP moistened in saliva contacts the tooth surface, the protective barrier breaks down, facilitating availability of calcium, phosphate and fluoride to the tooth; subsequently, enhancing remineralization.

Dicalcium phosphate dihydrate (DCPD) slurries are effective in preventing plaque pH drop and allowing higher degree of saturation for an extended period of time.

d. *NovaMin (Bioactive Glass)*

NovaMin is a bioactive glass ceramic (calcium sodium phosphosilicate) material, when exposed to aqueous medium, provides calcium and phosphate ions that form hydroxy-carbonate apatite (HCA) with time. NovaMin is able to fill the small surface defects in enamel and minimize development of erosion from acidic foods and beverages; also allows remineralization of these defects. NovaMin has been incorporated into dentifrices/gels (oravive revitalizing paste). A few studies have observed low availability of calcium and phosphate from NovaMin.

e. *Enamelon*

Enamelon consists of unstabilized calcium and phosphate salts with sodium fluoride. Enamelon toothpastes have shown promising results in reduction of white spot lesions and remineralization of tooth enamel.

A technical problem with Enamelon is that calcium and phosphate are not stabilized, allowing these two ions to combine into insoluble precipitates before coming in contact with oral fluids and enamel.

f. *Xylitol (Sugar Alcohol)*

Xylitol (sugar alcohol), used as a substitute for sugar, occurs naturally in plants.

Xylitol has the ability to:

- Make plaque less adhesive
- Reduce *S. mutans* levels
- Reduce plaque formation
- Neutralize plaque acids by decreasing the production of lactic acid

- Reduce caries progression
- Encourage remineralization

Xylitol stimulates salivary flow. It has shown promising results in combination with fluorides. Xylitol helps *S. mutans* bind to the proteins. Because the bacteria are bound, they are unable to produce sticky extracellular polysaccharides that bind microflora together; subsequently, less adhesion of bacteria to surface enamel.

Other sugar substitutes, such as sorbitol and isomalt, have also shown promising results.

g. *Ozone*

Ozone (three oxygen atoms) has effectively stimulated remineralization of incipient caries following treatment for a period of 6–8 weeks.

h. *Cold Plasma and/or Bioglass*

Cold plasma (non-thermal plasma) can effectively increase the surface hydrophilicity of dental substrate. Applying plasma alone to the substrate may not result in successful remineralization (result in deposition of porous layer of calcium phosphate on enamel surface). Application of bioglass after treatment with cold plasma increases the remineralizing effect. It has been established that bioglass can remineralize enamel with loss of around 100 µm depth; combining with application of cold plasma substantially increases its effectiveness.

i. *Ion-exchange Resins*

Ion-exchange resins (IER) have been used in pharmaceuticals as drug delivery systems. Toothpastes containing mixture of IER, which supply calcium, fluoride, phosphate and zinc ions promoting remineralization, are under research.

B. Remineralization of Dentinal Lesions

As the carious lesion reaches the dentin matrix, it progresses rapidly than enamel lesion. The dentinal lesions show different zones of mineral contents and optical reflections. These lesions are remineralized by spontaneous incorporation of calcium, phosphate and fluoride ions from oral fluids or by external sources.

Remineralization of dentin implies precipitation of mineral between collagen fibrils. This precipitation of mineral into demineralized dentin matrix may provide an increased minimal content, but do not chemically interact with organic compounds of the matrix.

The overall mechanical properties do not change. The preservation and stability of dentin collagen is essential during remineralization process, because it acts as scaffold for mineral deposition.

Plant extracts, polyphenols, green seed extract, contain proanthocyanidin (potent antioxidant and anti-inflammatory), etc. are used as dentinal remineralizing agents. These substances accelerate the conversion of soluble collagen to insoluble collagen during collagen synthesis. Polyanthrocyanidin also inhibits the enzymatic activity of glycol transferase; subsequently, inhibits caries progression.

V. RESTORING CAVITATED LESIONS: MINIMAL INVASIVE APPROACH

It is established that by focusing on infection control measures coupled with application of remineralizing agents, the restoration placement can be reduced to approximately 50%.

As the clinician detect a white spot lesion, efforts should be to educate and motivate the patient regarding its contributing factors and also the protocol for control. Uncontrolled demineralization leads to cavitation, which further encourages growth of oral microflora (mainly facultative anaerobes). Restoration should be avoided, especially in following situations, which may show evidence of dental caries:

- White spot lesions or hidden lesions, which are non-cavitated (especially when confined to enamel).
- Secondary caries along the restoration-tooth interface; preferably be managed using topical remineralizing agent.
- Root surface lesion, even if cavitated, should be managed using topical remineralizing agents (Inaccessible lesions do not fall under this category).

A classification system, termed ICDAS II (International Caries Detection and Assessment System) can be applied to individual tooth surfaces (Table 28.1).

ICDAS II system distinguishes between sound teeth surface, first visual change in enamel (white spot), distinct change in enamel (white lesion), microcavitation and cavitation of different depths. (Details of ICDAS, explained in author's book: *Dental Caries*—Vimal K Sikri.)

Cavitation makes plaque control difficult especially at the affected site; therefore, surgical approach is more reliable. The newer techniques of tooth preparation allow much more conservative preparation designs than those used earlier.

The classification conceived by Mount and Hume (Chapter 2) based on site and size of the lesion is considered the basis for minimal cavity preparation, though the advanced and extensive lesions do not fall under such regimes. Secondly, the classification is subjective; the size of the lesion is not well defined.

The tooth can be restored by one of the following modalities utilizing minimal invasive approach.

Table 28.1: Scoring criteria for ICDAS-II

Scoring	*Criteria*
0	Sound tooth surface: No evidence of caries after prolonged air drying (5 s).
1	First visual change in enamel: Opacity or discoloration (white or brown) is visible at the entrance of the pit or fissure after prolonged air drying, which is not or hardly seen on a wet surface.
2	Distinct visual change in enamel: Opacity or discoloration distinctly visible at the entrance of the pit and fissure when wet, lesion must still be visible when dry.
3	Localized enamel breakdown due to caries with no visible dentin or underlying shadow: Opacity or discoloration wider than the natural fissure/fossa when wet and after prolonged air drying.
4	Underlying dark shadow from dentin +/– localized enamel breakdown.
5	Distinct cavity with visible dentin: Visual evidence of demineralization and dentin exposed.
6	Extensive distinct cavity with visible dentin and more than half of the surface involved.
Caries associated with restoration and sealants	
0	Sound tooth surface with restoration and sealant
1	First visual change in enamel
2	Distinct visual change in enamel/dentin adjacent to restoration/sealant margin
3	Carious defect of >0.5 mm (with signs of code-2)
4	Marginal caries in enamel/dentin/cementum adjacent to restoration/sealant, with underlying dark shadow from dentin
5	Distinct cavity adjacent to enamel/dentin
6	Extensive distinct cavity with visible dentin

a. Use of hand instruments: Atraumatic restorative treatment (ART)
b. Use of rotary instruments
   - Tunnel preparation
   - Fissurotomy
   - Minibox preparation
c. Air abrasion
d. Chemo-mechanical caries excavation (preparation)
e. Ultrasonic excavation (preparation)
f. Fluorescence UV light
g. Laser cavity preparation

a. Use of Hand Instruments: Atraumatic Restorative Treatment (ART)

If a restoration is required for cavitated lesion, then every effort is made to conserve the tooth structure, making the restoration as small as possible. Adhesive restorative materials have made the task easier. ART technique involves the removal of affected tooth tissues with hand instruments, followed by restoration of the cavities with a specially designed glass-ionomer restorative material (GC Fuji IX) (Fig. 28.2).



Fig. 28.2: GC Fuji IX (powder and liquid)

The ART approach to manage caries lesions was first tried in the mid 1980s. Single surface restorations have shown better success rate than multisurface restorations.

Later, a few authors observed no difference between restoration survival for children and adults; however, marked operator effects were observed on the survival rate of restorations.

Procedure

The steps followed in atraumatic restorative treatment (ART) are:

- The concerned tooth/lesion is examined clinically as regard extent of demineralization (level of progress of caries into enamel at different depths) (Fig. 28.3A). The tooth is cleaned and isolated with sterilized cotton rolls.
- A dental hatchet or sharp excavator is used to gain access and excavate caries, if any (Fig. 28.3B). If caries do not exist, excavators may not be used.
- Undermined enamel is broken off with hand instruments and the soft dentin is excavated (Fig. 28.3C).
- The prepared cavity is cleaned using appropriate solutions (Fig. 28.3D). In case of dentinal caries, there is no danger of leaving unsupported enamel (after removing dental caries), since the undermined dentin when restored with GIC, become 'supported'. In routine, ART preparations do not need lining/base material.
- Dentin conditioners, may be weak polyacrylic acid, is used to remove smear layers (Fig. 28.3E). In case the conditioned tooth surface becomes contaminated, the preparation is washed, dried and re-conditioned. Liquid of GIC (not of water settable GIC) is an effective conditioner of both enamel and dentin.
- Glass-ionomer cement is filled in the cavity and also in any confluent pits and fissures (Fig. 28.3F).

Advantages

- No sophisticated dental equipment is needed (can be used in rural areas).
- Treatment is not dependent on electricity.
- Minimal discomfort to the patient.
- Operator requires minimal training.
- Fluoride release from glass-ionomer cement will have a cariostatic effect.
- Cost is very low.
- If this treatment fails, a more radical approach may be used.
- Less time consuming.

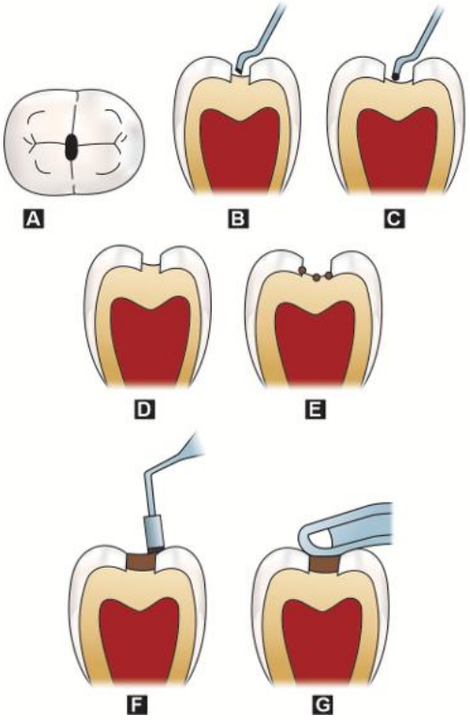


Fig. 28.3: ART approach: **(A)** Evaluating extent of lesion; **(B)** Gaining access with sharp instrument; **(C)** Excavating soft dentin; **(D)** Cleaning of prepared cavity; **(E)** Conditioning with weak polyacrylic acid; **(F)** Filling glass-ionomer cement, **(G)** Pressing filling with vaseline-coated finger

- Hand instrument is used to finish the restoration. The finished restoration is covered with petroleum jelly or varnish. Vaseline-coated finger is pressed over the restoration (Fig. 28.3G).

Hall Technique

Hall technique is a newer method for managing carious lesions. Preformed metallic crowns are used to seal decay by cementation over carious primary molars with a glass-ionomer luting cement. Tooth preparation and caries excavation are not required for this technique.

b. Use of Rotary Instruments

i. *Tunnel Preparation*

The tunnel preparation is performed by accessing the proximal carious lesion from the occlusal surface, while preserving the marginal ridge.

When caries is present only on proximal side, usually the access is gained through occlusal side. Conventionally, after making occlusal dovetail and breaking the marginal ridge, proximal box is prepared.

Such restorations lead to more of tooth loss and jeopardize the integrity of marginal ridges. In tunnel preparations, the access is gained through occlusal side without involving the marginal ridge.

Tunnel restorations were designed with the idea of preserving the marginal ridge during posterior proximal restorations. For convenience, the tunnel restorations are divided into three types, viz. internal, partial, and total (Fig. 28.4). The 'internal tunnel' implies touching the lesion; the main objective is to help remineralization. This is also known as 'terminated tunnel' preparation. In partial tunnel preparation, the enamel is smoothened around the periphery and the remaining lesion can remineralize. In 'total tunnel' preparation, all demineralized enamel is removed.

The adjacent tooth is protected with matrix bands before cavity preparation. The access is gained through the mesial/distal pit depending upon the side involved. A small round bur is pierced through the pit, directing the bur at 45° towards the proximal contact area (Fig. 28.5A, B). Once the lesion is approached, the access can be slightly widened using tapering fissure burs. The carious lesion is thoroughly excavated by moving the bur in occlusal and gingival directions. Till the completion of the cavity preparation and the restoration, the band is kept stable so as not to injure the adjacent tooth and the underlying gingiva. Various studies conducted on tunnel restorations could not show promising results. The strength of the marginal ridge is not maintained. The chances of leakage and secondary caries are more with

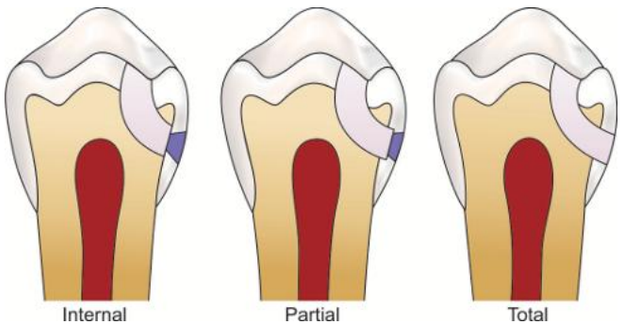


Fig. 28.4: Tunnel preparation

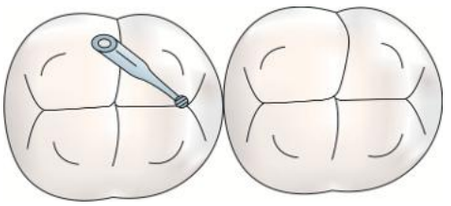


Fig. 28.5A: Direction of the bur (occlusal view)

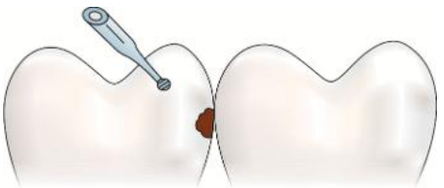


Fig. 28.5B: Direction of the bur (cross-sectional view)

these preparations. However, these are indicated in small lesions, especially in deciduous teeth.

In case marginal ridge is weakened, the same is involved in cavity preparation; only proximal box is involved (Refer to as Minibox preparation).

ii. *Fissurotomy*

Fissurotomy is a new modality utilizing ultra-conservative approach in cavity preparation. The fissurotomy kit contains fissurotomy burs, acid etchant, bonding agent and flowable composite (Glass-ionomer cement can also be used.) The tip size of fissurotomy bur is ten times smaller than the conventional #556 carbide bur. The bur creates narrow and round internal line angles.

The characteristics of fissurotomy bur are:

- *Shape:* Tapering
- *Head length:* 2.5 mm
- *Bur tip:* Ten times smaller than # 556 carbide bur
- Bur produce rounded line angles (desired in composites)
- Tip controls the depth and width of preparation

Clinical steps

- Probe pits and fissure areas; explore for decay and morphological features
- Gently contour the surrounding enamel
- Width of restoration, as minimum as possible (say 1/8th to 1/10th intercuspal distance)
- Remaining caries, if any, to be removed (Caries detector dye can be used.)
- Wash the preparation thoroughly
- Final preparation would be narrow and may be irregularly deep
- Restore with flowable composite/glass-ionomer cement

Advantages
• Short head length of the bur allows the operator to control depth of the preparation • Shape of the bur facilitate conservative preparation • Minimum heat build-up • No vibrations (Patient discomfort is minimized.) • No need for local anesthesia

iii. *Minibox Preparation*

Minibox preparation involves the removal of the marginal ridge, but does not include the occlusal pits and fissures, if caries removal in these areas is not necessary. These cavities can be restored with composite or bonded amalgam (Fig. 28.6).

Advantages
• Preservation of tooth structure • Maintenance of marginal ridge and proximal contacts • Avoidance of iatrogenic damage to the adjacent tooth during cavity preparation • Fluoride supply to adjacent tooth • Negligible effect on gingival tissue • Cost-effective • Esthetic appearance • Risk of overhang is minimal

Disadvantages
• Limited accessibility • Risk of incomplete removal of caries • Marginal ridge being undermined, may break • Risk of pulpal involvement • Marginal adaptability of the restoration is poor • Difficulty in insertion and finishing of the restorative material

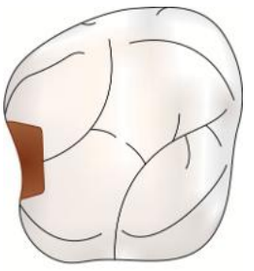


Fig. 28.6: Minibox preparation

c. Air Abrasion

The air abrasive technique might not be effective as an alternative to rotary instruments; however, this can be utilized as a supplement in the treatment of caries.

Air abrasion is non-rotary method of cutting hard tissues. The technique uses pseudo-mechanical kinetic energy from a stream of abrasive particles thrown at

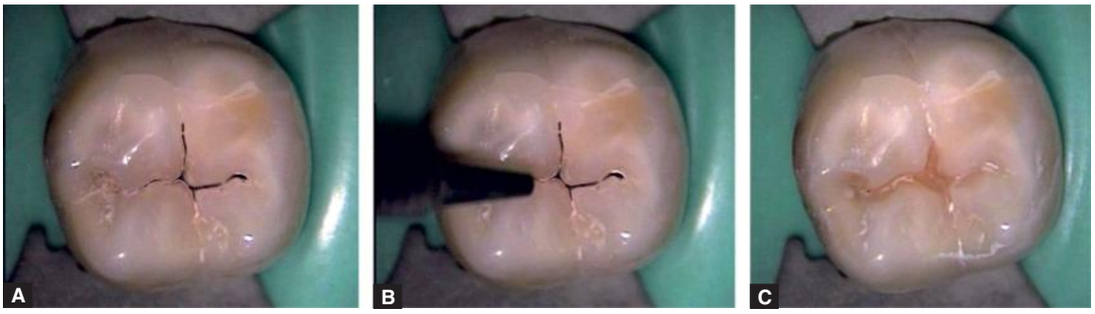


Fig. 28.7: Air abrasion cavity preparation

the tooth surface at a certain velocity. Kinetic energy can be increased by increasing the velocity or increasing the mass (*Rotary instrumentation is mechanical energy; whereas, air abrasion is kinetic energy*).

The commonly used abrasive particles are aluminum oxide with an average particle size of 25–30 µm. Recently, bioactive glass is being used in place of aluminum oxide. The particle size varies with the requirement of the cavity design and the morphology of the tooth. The normal exit pressure utilized is between 60 and 120 pound per square inch. The velocity of the particle exit will depend upon the diameter of the nozzle tip. The particles escape from the tip in a conical manner; the height and width of the cone depends upon the size of the nozzle. The dimension also increases as the distance between nozzle tip and the tooth increases. The cavity preparation can be precise, if the tip is placed near the tooth surface (Fig. 28.7A to C). The internal line angles are usually rounded by air abrasive technique; however, if the tip is placed within 1.0 mm from the tooth surface, the line angles can be sharp. In case when beveled preparations are required, the nozzle tip is kept at a distance of 5.0 mm. For the preparations requiring a butt joint, the tip is placed 1.0 mm from the tooth surface. Since air abrasion follows end cutting pattern, the depth of the cavity can be more than required. The size and type of the particles, speed, distance and length of time play an important role in the cutting efficiency.

Indications

- Removal of superficial enamel defects
- Detection of pit and fissure caries; removal of organic debris facilitates detection
- Removal of superficial enamel stains
- Removal of minimal caries lesion
- Surface preparation of abfraction and abrasion lesions (remove the glaze for better bonding)
- Removal of existing restorations
- No need for anesthesia (cooling effect of the high pressure air provide comforts)

Contraindications

- Chronic pulmonary diseases, viz. asthma patients
- Severe dust allergy
- Teeth adjoining recent extraction
- Subgingival caries removal

Advantages

- Less invasive procedure; preserves more tooth structure
- Non-traumatic
- Can remove old restorations and/or cements
- No shattering of enamel or microfracture as with conventional bur
- Rare need of anesthesia
- Technique is fast and silent
- No heat is produced (less discomfort to the patient)

Disadvantages

- Rubber dam, high velocity evacuation and protective eyewear are mandatory
- Lack of tactile sensation
- Carious removal may be inadequate (non-contact based modality)
- Aluminum oxide particles may spread around the dental operatory (Splattering of powder is a great risk of ingestion)
- Impaired access (Abrasive particles easily collect on the mirror and block the viewing surfaces).
- May remove normal dentin
- Remove exposed cementum and root dentin
- Well-defined cavity margins cannot be prepared
- Does not obviate the need for acid etching
- May lacerate soft tissues
- Removal of amalgam restoration may cause release of mercury in the environment
- Not cost-effective

Uses

- *Minimal cavity preparation:* It requires minimum cavity preparation (The cavity is at the requisite location).

- *As an adjunct to acid etching:* It is hypothesized that abraded enamel surfaces are irregular, the pattern dissimilar to the acid-etched enamel. The air abrasion alone cannot be used as a substitute of acid etching. Air-abraded dentin shows only smear layer with no open dentinal tubules. It is established that the use of air abrasion along with acid etching provides good results.
- *As a repair modality:* The repair of fractured restorations poses difficulty for the operator. The composite to porcelain union is enhanced using hydrofluoric acid. It is established that use of air abrasion followed by application of hydrofluoric acid provides much better union. In case of composite repair, the contamination of surface layer make it difficult to achieve bonding at the repair site. The silicate ceramic particles with and without alumina particles have shown better results in providing the necessary roughened surfaces. A few authors, however, are of the view that such roughening does not provide better bonding during composite-to-composite repair.
- *As a preventive measure:* Air abrasion technique has been tried in preventive measures; facilitates application of pit and fissure sealants and also small tunnel preparations.
- *Detecting early carious lesions:* The air abrasive technique has been used in detecting the early carious lesions, mostly pit and fissure caries. For detecting caries, air polishing using sodium bicarbonate powder is utilized to remove the surface stains, plaque, etc. The sodium bicarbonate powder does not disturb the enamel surfaces. The tooth can be washed, dried and examined for any opacity on the surface of enamel. Air abrasion with routine alumina powder is avoided in detecting caries, as alumina abrasive can damage the surface and lead to microscopic cracks onto the enamel surface.

Air Abrasion (Kinetic Cavity Preparation)

- Air abrasion uses a pressurized stream of small aluminum oxide particles to abrade carious lesion / enamel surface.
- Air abrasion system includes:
  - Small particles (diameter 25–50 μm)
  - Adequate powder flow rate (0.7–4.2 g/min)
  - Air pressure (40–160 psi)
  - Operating distance from tooth (tip placement: 1.0–2.0 mm)

d. Chemo-Mechanical Caries Excavation (Preparation)

The conventional caries excavation methods usually induce pain and fear in patients, especially the children. To make the caries removal painless, selectively soften the carious lesion by various chemical agents (Fig. 28.8) facilitates the removal by excavation without any pressure.



Fig. 28.8: Chemical agents used in softening carious lesion

The chemical agents used are as follows:

i. *Sodium Hypochlorite-based Agents*

Earlier authors used 5.0% sodium hypochlorite to soften carious lesions. Later, the use of 5.0% sodium hypochlorite was observed to be toxic to adjacent healthy tissues. A new solution, GK-101 was developed, which consisted of sodium hydroxide, sodium chloride and glycine in addition to 5.0% sodium hypochlorite. GK-101 was more effective than hypochlorite alone; however, the carious softening effect was very slow. Caridex, a commercial preparation, consists of sodium hypochlorite solution buffered with an amino acid containing mixture of aminobutyric acid, sodium chloride and sodium hydroxide. The ability of sodium hypochlorite to selectively soften carious lesion is attributed to the buffering effect of amino acid, which enhances disrupting effect on the degenerated collagen of the carious dentin. The resultant friable collagen fibrils can be easily removed with excavators. Caridex was not fully successful in clinical practice, because of the need of a specific apparatus to deliver the solution into the cavity, short shelf-life, longer treatment time, and higher treatment cost. Carisolv is an another caries removing agent based on sodium hypochlorite. It consists of two components; one is liquid containing 0.5% sodium hypochlorite and the other component contains an amino acid mixture (glutamic acid, leucine and lysine), sodium carboxymethyl cellulose (to enhance the viscosity), sodium hydroxide (provides pH of 11) and water to act as vehicle. The chemical

composition and microstructure of dentin after excavation with Carisolv remained unchanged. The calcium and phosphorus contents remain similar after excavation; however, the hardness values of residual dentin was same as for sound dentin. The etch pattern of dentin in Carisolv-excavated lesions is deeper than caries removed with a conventional bur. Carisolv excavated dentin shows better bonding characteristics, viz. patent dentinal tubules (not covered by smear layer), irregular surface and the improved wetting potential. Carisolv has superior property in reducing the counts of viable bacteria in residual dentin, as compared to conventional bur excavation. Carisolv has bactericidal activity due to formation of chloramine compounds. The disadvantages include its high cost and need of special instruments. Children usually dislike the chlorine taste and odor. The procedure of excavation of caries using Carisolv is depicted in Fig. 28.9A to C. The process is repeated until no turbidity of the applied gel is seen.

The cavity preparation using carisolv includes:

a. Local anesthesia is not required; however, it can be administered, if patient is apprehensive.
b. Enamel lesion can be removed with small round bur. The agent is applied to the carious dentin.
c. After waiting for 30 seconds, the caries is excavated using excavators. The lesion is kept soaked with the gel and during subsequent application, no waiting period is required.

ii. *Pepsin-based Caries Excavation*

An enzyme pepsin in a phosphoric acid/sodium bisphosphate buffer has also been tried as a chemical agent softening caries. SFCVIII (3M ESPE) is the commercially available pepsin-based agent. The advantage of this new pepsin-based agent is that it acts on more specific denatured collagen than the sodium hypochlorite-based agents. The phosphoric acid dissolves the inorganic component of carious dentin, while at the same time pepsin gets access to the organic part of the carious mass, selectively dissolving the denatured collagen. SFC-VIII gel should be used to remove caries along with a excavator having hardness between that of sound and infected dentin. The limitation of SFC-VIII excavation is that the heavily pigmented and arrested dentin caries pose problem in pepsin digestion.

Advantages
• Painless, comfortable for the patients • Conservative cavity preparation • Patient's acceptance

Disadvantages
• Time consuming • Rotary instruments may be required

Papacarie

Papacarie (Sao Paulo-Brazil) is another chemo-mechanical caries removing agent, which contains papain, chloramines, toluidine blue and certain salts. Papain is a proteolytic enzyme similar to pepsin. Papain comes from latex of leaves and fruits of green adult papaya (Carica papaya). Papacarie is available in syringes having 3.0 mL gel. The papacarie gel is allowed to act in a cavity for 60 seconds. As the collagen fibrils get dissolved, the clear gel becomes dark in color. The papain digests dead cells and breaks the partially degraded collagen molecules, contributing to the degradation of collagen fibrils formed by the carious process. The procedure of excavation of caries using Papacarie is depicted in Fig. 28.10A to C.

Carie-Care

It is an enzyme derived from papaya plant, 'the Carica papaya'. It is composed of papaya extract, therapeutic essential oil, coloring gel, sterile water, chloramines and sodium chloride. Carie-Care (papain) acts on the carious tissue, which lacks alpha-antitrypsin (an anti-protease system), which inhibits proteolysis. The absence of antiprotease allows papain to act on the partially degraded collagen by breaking the peptide bonds. In addition to papain, the chloramines dissolve carious dentin by means of chlorination of the



Fig. 28.9: Removal of caries using Carisolv: **(A)** Making access cavity; **(B)** Applying Carisolv on the caries lesion; **(C)** Removing the turbid material using spoon excavator

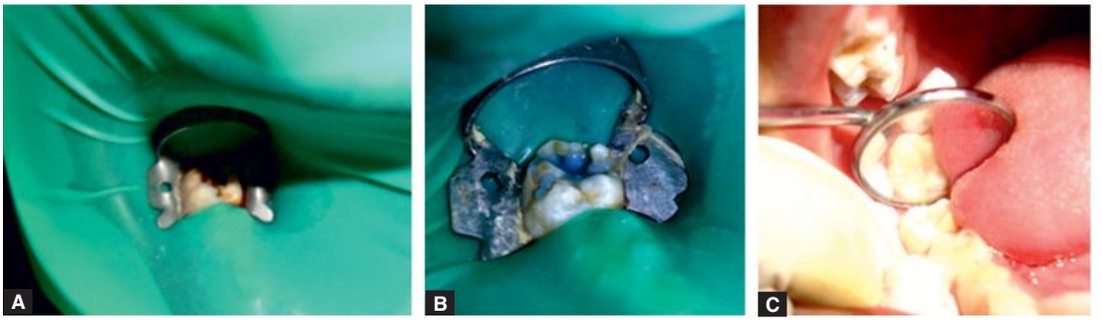


Fig. 28.10: Removal of caries using Papacarie: **(A)** Rubber dam application; **(B)** Application of Papacarie; **(C)** Caries removed

partially degraded collagen. The gel also contains therapeutic oils (clove oil), which induce analgesic and antiseptic action. It minimizes pain sensation and has a pleasant taste acceptable by the children and adults as well.

e. Ultrasonic Excavation (Preparation)

Ultrasonic energy with a 25 kHz oscillating frequency has been tried in cavity preparation. It is used with thick aluminum oxide and water slurry, resulting in the cutting action of tooth. The kinetic energy of water molecules is transferred to the tooth surface through high speed oscillations of the cutting tip. The disadvantage of ultrasonic excavation is that it is unable to remove soft, carious dentin; whereas, the leathery, deeper layers are more susceptible to ultrasonic action.

Excavation by Sono-Abrasion

It is based on the use of ultrasonic cutting tips along with sonic air-scaler handpieces under water cooling. The handpiece oscillates in the sonic region (<6.5 kHz), while the tips perform an elliptical motion. A requisite torque is mandatory, otherwise the cutting efficiency is reduced. The advantage of this technique is that no chemical or structural changes were observed in dentin. Sono-abrasion excavation using diamond-coated tips appeared as efficient as conventional hand excavation using excavators. Cariex system (Kavo Dental, Germany) includes two sets of cutting tips; two diamond-coated tips with different diameters for enamel preparation and two tungsten carbide tips with different diameters for dentin excavation.

The advantages and disadvantages of sonic and ultrasonic preparation of Class II lesions are summarized in Table 28.2.

Advantages

- Safe and efficient treatment modality for caries removal
- No need for anesthesia, thus avoiding any numbness
- No vibration, little noise
- No smell

Disadvantages

- Cost factor
- Need to learn the technique

f. Fluorescence UV Light

During caries removal, UV light causes porphyrins inside caries to fluoresce orange/red color; whereas, the surrounding healthy tooth structure fluoresces light green. The color variation enables the operator to excavate only the caries part, leaving the healthy tissue intact. This device is also helpful in detecting caries even in early stages. The operator wears yellow tinted glasses to view caries or yellow tint is inserted in the loupes. The glasses are provided with the fluorescence UV kit.

Table 28.2: Sonic and ultrasonic minimal invasive preparation in Class II: Advantages and disadvantages

Preparation regime	*Advantages*	*Disadvantages*
Ultrasonic	• Preservation of the adjacent proximal surface • Preservation of the marginal ridge (slot and tunnel cavity) • Minimal preparation (esthetics preserved)	• Ultrasonic vibrations may lead to microcracks • Difficult to control vibrations
Sonic	• Preservation of the adjacent proximal surface • Preservation of the marginal ridge • Minimum vibrations • Minimal preparation (esthetics preserved)	• High air pressure required • May be time consuming (depends upon hardness of tooth tissues)

g. Laser Cavity Preparation

Lasers are being widely used for cavity preparation. The Er:YAG and Nd:YAG lasers have been used to cut dental hard tissues; whereas, CO_2 lasers facilitate sealing of fissures. Laser treatment reduces the number of carious bacteria and also volatilizes the water in the caries. With the use of lasers, the effect of vibrations, pressure and unfavorable temperature changes associated with the use of rotary cutting instruments is eliminated. The cavity cutting by lasers is clean with minimum debris and smear layer. The morphology of enamel prisms and dentinal tubule is preserved. Lasers facilitate selection removal of caries. Preparations are similar to those made with air abrasion. Adhesive materials are preferred for restoration.

The erbium yttrium-aluminum-garnet (Er:YAG—wavelength 2.78 mm) and the erbium-chromium—yttrium-scandium-gallium-garnet (Er,Cr:YSGG-wavelength 2.94 mm) lasers are currently used in cavity preparation and caries excavation. The mechanism by which enamel and dentin are removed consists of explosive subsurface expansion of water interstitially trapped in the dental hard tissues. During irradiation, the water molecules absorb the incident radiation, causing sudden heating and water evaporation. As a result, a high stream pressure is formed, inducing a violent expansion and ejection of dental hard tissues. Er, Cr:YSGG laser system delivers photons through air-water spray straight onto the target tissue. The microexplosive forces turning into water droplets, contribute to the hard-tissue removal. Laser ablation provides more comfort to the patient due to the absence of vibration and a lower pain sensation. It has definite role in the prevention of recurrent caries.

Laser cavity preparation: Major drawbacks

- Time consuming; the time required for a complete excavation is twice that with rotary instruments.
- Can induce irreversible chemical and structural alteration of dentin and even damage the pulp.
- Produces an undefined and irregular excavation pattern in dentin.
- Irregular dentin surface left after laser ablation hinders tactile sensation.
- Laser may change the organic matrix component of dentin, thereby impairing the adhesive treatment.

Minimal invasive techniques, without preparation and with minimal preparation are summarized in Table 28.3. The advantages and disadvantages of conditioning agents are summarized in Table 28.4.

Table 28.3: Minimal invasive techniques: Tissue conditioning (without preparation) and with minimal preparation

Surface involved	*Tissue conditioning (without preparation)*	*Tissue conditioning (Minimal preparation)*
Occlusal/buccal/lingual or palatal	• Fluoride and chlorhexidine varnish • Glass-ionomer varnish and GC Tooth Mousse • Ozone therapy • Remineralizing solutions • Resin infiltration	• Minimal preparation with air abrasion, laser, etc.
Proximal surface	• Fluoride and chlorhexidine varnish • Glass-ionomer varnish and GC Tooth Mousse • Resin infiltration	• Minimal preparation with air abrasion, etc. • Tunnel cavities with icon infiltration.

Table 28.4: Advantages and disadvantages of conditioning agents

Surface involved	*Tissue conditioning agents*	*Advantages*	*Disadvantages*
Occlusal/buccal/lingual or palatal	• Fluoride varnish	• Easy application	• Efficiency depends upon caries risk assessment
	• Glass-ionomer and GC Tooth Mousse	• Easy application	• Efficiency depends upon caries risk assessment
	• Ozone therapy	• Esthetically better	• Efficiency doubtful
	• Resin infiltration	• Better penetration	• Action can be reversible
Proximal surface	• Fluoride varnish	• Application difficult	• Efficiency depends upon caries risk assessment
	• Glass-ionomer and GC Tooth Mousse	• Application difficult	• Efficiency depends upon caries risk assessment
	• Resin infiltration	• Better penetration	• Long-term efficiency not confirmed

Table 28.5: Management of carious substrate: Minimal invasive techniques

Mechanism	*Technique*
Mechanical (rotary)	Burs/diamond points
Mechanical (non-rotary)	Hand instruments, air-abrasion ultrasonics, sono-abrasion
Chemo-mechanical	Caridex, Carisolv, Pepsin-based solutions/gels
Photoablation	Lasers
Disinfection	Photo-active disinfection, ozone

The minimal invasive techniques managing the carious lesions are summarized in Table 28.5.

The relative ability of excavating modalities to remove caries and tooth tissue is summarized in Table 28.6.

Restorative Materials

Adhesive dental materials make it possible to conserve tooth structure, because they do not require the inclusion of mechanical retention features. Examples of such materials are glass-ionomer cements, resin-based glass-ionomer cement, composite/dentin bonding agents, and the bonded amalgam.

i. *Glass-ionomer Cements*

The advantages of glass-ionomer cement (GIC) include, adhesion to tooth structure and release of fluorides. They perform well in low stress areas. Set GIC is 'rechargeable'; can take up fluoride from the environment, which is provided by exposure to fluoride varnishes and fluoride containing toothpastes.

The disadvantages include, technique sensitivity and the opaque nature. The handling properties and opaqueness of the material can be overcome by adding resin to the material resulting in resin-modified GICs, which are easier to place, are light cured, and have improved esthetics.

ii. *Composite Resins*

The cavity preparation for composite mainly depends upon the surface area of the tooth, which can be utilized for bonding. Cavity preparations designed to conserve maximum enamel can eliminate the need for macromechanical retention. Bonding, both to dentin and enamel is required for composite restorations. Various types of composites are available for use in anterior and posterior teeth. Flowable composites with lower filler loading have been developed, which are preferred in inaccessible areas. These materials have low modulus of elasticity and higher polymerization shrinkage, therefore, not used in routine.

iii. *Bonded Silver Amalgam*

The bonded silver amalgam was introduced with an idea to improve the bonding of amalgam to the cavity walls. A chemically active, resin-based luting material is placed along the cavity wall and at its base before condensing the silver. Though the adhesion is definitely improved, the other physical properties might be compromised, since proper depth required for silver is not provided. The thickness of the oxygen inhibited layer in dentin bonding agents may create difficulty in bonded amalgam.

Advantages

- Amalgam can be used in short clinical crowns.
- Marginal leakage is checked.
- Conservative approach is maintained.
- Sealed and bonded restorations prevent further progress of caries.

Disadvantages

- Increased cost
- Proper strength may not be achieved.

Table 28.6: Minimal invasive techniques: Relative ability of excavating modalities to remove caries and tooth tissue

Modality	*Sound enamel*	*Enamel caries*	*Sound dentin*	*Dentin caries*
Hand excavator/hatchet	Nil	Good	Nil	Very good
Rotary burs (metallurgy/configuration is important)	Excellent	Excellent	Excellent	Excellent
Air-abrasion (abrasive technique/agent is important)	Excellent	Very good	Excellent	Good
Sono-abrasion (under initial stages)	Nil	Good	Good	Very good
Ultrasonic (mainly for retrograde preparation)	Good	Good	Good	Nil
Caridex/Carisolv (adequate access is important)	Nil	Nil	Nil	Excellent
Lasers (type of laser is important)	Good	Good	Good	Good
Enzymes (under developing stage)	Nil	Nil	Nil	Good

VI. CARIES INHIBITION BY RESIN INFILTRATION

Resin infiltration is a microinvasive treatment regime for arresting early carious lesions. The initial carious lesion is porous due to loss of minerals from the tooth surface.

These porosities can be infiltrated by low viscosity resin [A low viscosity resin with a high penetration coefficient has been developed by Dental Milestones Guaranteed (DMG) and is marketed as ICON caries infiltrant.], which lead to strengthening of the demineralized tooth surface without any drilling. The infiltration procedure also provides optical blending of the carious lesion with the healthy enamel. The concept of resin infiltration was first developed at the University of Kiel (Germany) to be used for the management of smooth surface caries and proximal non-cavitated carious lesions.

Indications

- White spot lesions (early caries)
  - Proximal
  - Cervical
  - Occlusal
- Fluorosis stains (non-pitted)
- Amelogenesis imperfecta
- Molar-incisor hypomineralization

The types of white spot lesions, their etiology and treatment options are summarized in Table 28.7.

Requirements of Resin Infiltrant

The requisite characteristics of resin infiltration are:

- Should be hydrophilic, highly surface active and with low viscosity
- Bacteriostatic
- Non-toxic to oral tissues
- Easily polymerizable in oral environment
- Should resist chemical and mechanical challenges of the oral cavity
- Esthetically pleasing

Resin Infiltration Kit

The resin infiltration kit is available separately for vestibular and proximal lesions containing three syringes: (i) Acid gel (ICON Etch), (ii) drying agent (ICON Dry), and (iii) resin infiltrant (ICON Infiltrant) along with special applicator tips (Fig. 28.11). The Etchant (acid) gel is composed of 15% hydrochloric acid, water, silica, etc., which effectively etches the tooth surface. Drying agent is ethanol, which removes water retained in microporosities. The Infiltrant is composed of triethylene glycol dimethacrylate (TEGDMA), a light cure low viscosity resin, which penetrates the pores due to capillary action, filling the pores completely. The microporosities created by demineralization process and subsequently filled with resins have been evaluated. Even earlier studies have



Fig. 28.11: Resin infiltration kit

Table 28.7: White spots: Types, etiology and treatment option(s)

Types of white spots	*Etiology*	*Treatment option(s)*
• White spots	• Natural occurrence in maxillary incisors	• Whitening only
• Mottled enamel (white specks)	• Development (less chances)	• Whitening and microabrasion
• Brown/white discolorations	• Fluorosis (multiple lesions)	• Whitening and microabrasion
• White patches	• Trauma to the primary dentition	• Whitening, followed by resin infiltration
• White spots with yellowish covering	• Bleeding during traumatic injury, which might have seeped into the areas of mineralization	• Whitening and resin infiltration
• White lesions with black margins	• Demineralization lesions after removal of orthodontic brackets	• Resin infiltration/whitening/micro-abrasion, depending on the size
• Enamel defects in deciduous teeth	• Molar incisor hypoplasia	• Resin infiltration of the anterior lesions; GIC filling in posterior
• Enamel hypoplasia	• Developmental defect	• Whitening/microabrasion/resin infiltration

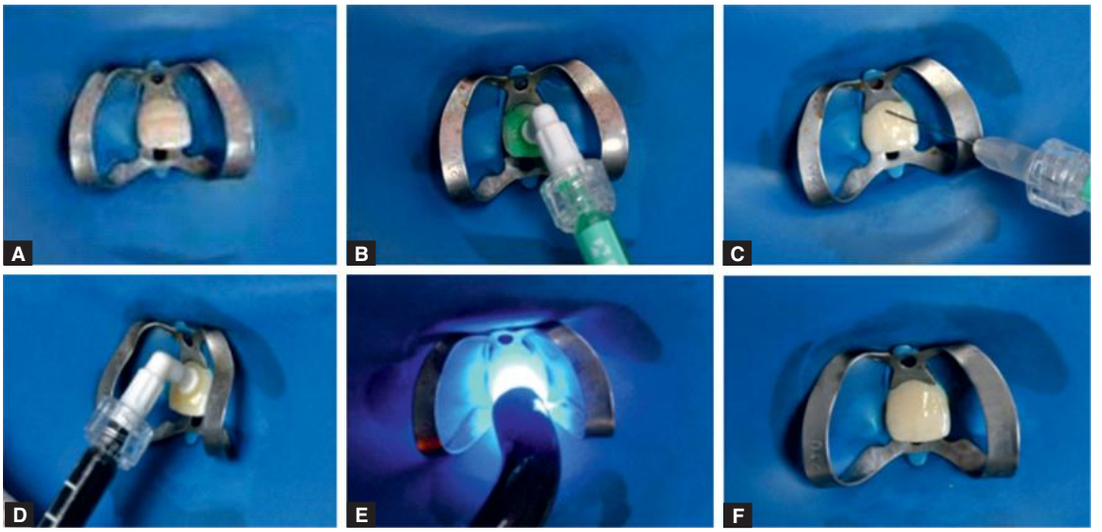


Fig. 28.12: Resin infiltration procedure: **(A)** Application of rubber dam; **(B)** Application of ICON Etchant; **(C)** Application of ICON Dry; **(D)** Application of ICON Infiltrant; **(E)** Light curing the Infiltrant; **(F)** Postoperative

demonstrated that the penetration of resins into demineralized surfaces inhibited the progress of early caries lesions; even though the resins available at that time might not penetrate effectively deep due to their high viscosity (Higher the viscosity lesser is the penetration coefficient).

Researchers continued to develop resins of low viscosity capable of penetrating deep into the pores created by demineralization. The resins exhibiting rapid penetration abilities were termed as 'Infiltrants'. The infiltration depth of these resins was evaluated clinically. The infiltration of 'low viscosity' resins into natural tooth surface was difficult; might be due to presence of pseudo-intact layer covering the porous structure hampering the infiltration. The surface layer is formed by precipitation of minerals on the outer surface and has a much lower pore volume compared to the lesion body, thus inhibiting penetration of the resin.

Conventional phosphoric acid etching was not effective as the penetration of resins remained superficial (phosphoric acid affected only 25 µm of surface). 15% hydrochloric acid for two minutes was tried as an alternative etching protocol. It removed 58 µm of demineralized/normal enamel surface. Etching with 15% HCl gel for 90–120 seconds was found to be more effective than 37% phosphoric acid gel (Hydrochloric acid in concentration of 18% has been used in dentistry for microabrasion). It is observed that the erosion depth with hydrochloric acid is more than twice as compared to phosphoric acid when etched for two minutes. HCl penetrates into deepest part of the lesion, preventing further attacks. The addition of ethanol after etchant application significantly increases the penetration of infiltrant due to decrease of viscosity and contact angle. Better penetration was observed using combination of TEGDMA and HEMA as infiltrant along with using 20% ethanol.

The clinical procedure of resin infiltration on smooth surface is depicted in Fig. 28.12A to F.

Advantages

- Improved esthetics
- Occlusion of microporosities (up to certain depth)
- Obturation of deeply porous demineralized areas
- Arrest/retardation of lesion progression
- Minimizes risk of secondary caries
- Mechanical stabilization of demineralized enamel
- Preservation of tooth tissue
- No risk of postoperative sensitivity
- Pulpal reactions are negligible.
- Restorative intervention can be delayed.
- Time and cost-effective
- Reduced risk of gingivitis and periodontitis

(For details of Resin Infiltration, refer author's book Dental Caries: Vimal K Sikri.)

VII. REPAIR RATHER THAN REPLACEMENT OF DEFECTIVE RESTORATIONS

The replacement of existing restoration because of varied reasons is a common feature. It is established that with every replacement, the loss of tooth structure increases. This loss is much more in case of composites and other tooth-colored materials. The repeated restoration replacement leads to the need for more complex treatments, may be indirect restorations. The

repair process, earlier considered as 'patchwork', is being accepted in the profession with the evolution of bonding agents and other adhesive materials.

The preparations should ensure independent retention and resistance form for the repair. The decision to repair rather than replace a restoration should be based on the patient's risk of developing caries, the professional's judgment of benefits vs. risks and conservative principles of cavity preparation.

The composite has shown promise in re-attachment of the fractured anterior teeth. The fracture of anterior crowns is very common in young age and the technique of re-attachment with composite produces requisite strength and adequate esthetics. The repair of fracture of ceramic crowns, both metal ceramic and all ceramic, has also been attempted using an abrasive silica impregnated technique. Repair with glass-ionomer cement may be preferred in cervical areas.

BIBLIOGRAPHY

1. Alonso RC, Correr GM, Borges AF, Kantovitz KR, Rontani RM. Minimally invasive dentistry: Bond strength of different sealants and filling materials to enamel. Oral Health Prev Dent 2005;3:87–91.
2. Ardu S, Perround R, Krejci I. Extended sealing of interproximal caries lesions. Quint Int 2006;37:423–27.
3. Bakry AS, Marghalani HY, Amin OA, Tagami J. The effect of a bioglass paste on enamel exposed to erosive challenge. J Dent 2014;42:1458–63.
4. Banerjee A, Watson TF. Air abrasion: its uses and abuses. Dent Update 2002;29:340–43.
5. Banerjee A, Thompson ID, Watson TF. Minimally invasive caries removal using bio-active glass air-abrasion. J Dent 2011;39:2–6.
6. Banerjee A, Watson TF, Kidd EAM. Dentin caries excavation: a review of current clinical techniques. Br Dent J 2000;188:476–80.
7. Banerjee A. Minimal invasive dentistry: part 7. Minimal invasive operative caries management: rationale and techniques. BDJ 2013;214:107–10.
8. Beelay JA, Yip HK, Stevenson AG. Chemomechanical caries removal: a review of the techniques and latest developments. BDJ 2000;188:427–31.
9. Bernardon JK, Gondo R, Baratieri LN. Minimal invasive restorative treatments of hypoplastic enamel in anterior teeth. Am J Esthet Dent 2011;1:10–14.
10. Burke FJ. Ozone and caries: a review of the literature. Dental Update 2012;39:271–74.
11. Burke FJT. From extension for prevention to prevention of extension: (Minimal Intervention Dentistry). Dent Update 2003;30:492–96.
12. Chen M, Zhang Y, Sky Driver M, Caruso AN, Yu Q, Wang Y. Surface modifications of several dental substrates by non-thermal, atmospheric plasma brush. Dent Mater 2013;29:871–80.
13. Christensen GJ. The advantages of minimally invasive dentistry. JADA 2005;136:1563–65.
14. Czarmecka B. The use of ART technique in modern dental practice: a personal view. J Dent 2006;34:620–22.
15. Eichenberger M, Biner N, Amato M, Lussi A, Perrin P. Effect of magnification on the precision of tooth preparation in dentistry. Oper Dent 2018;43:501–07.
16. Ericson D, Kidd E, McComb D, Mjor I, Noack MJ. Minimally invasive dentistry: Concepts and techniques in cariology. Oral Health Prev Dent 2003;1:59–72.
17. Featherstone JBD, Domejean S. Minimal invasive dentistry: part 1. From 'compulsive' restorative dentistry to rational therapeutic strategies. BDJ 2012;213:441–45.
18. Fontana M, Gonzalez-Cabezas C. Minimal intervention dentistry: part 2. Caries risk assessment in adults. Br Dent J 2012;213:447–51.
19. Frencken Jo E, Peters MC, Manton DJ, Leal SC, Gordan V, Eden E. Minimal intervention dentistry for managing dental caries-A review. Int Dent J 2012;62:223–26.
20. Guerrieri A, Gaucher C, Bonte E, Lasfargues JJ. Minimal intervention dentistry: part 4. Detection and diagnosis of initial caries lesions. Br Dent J 2012;213:551–57.
21. Holmgren CJ, Roux D, Domejean S. Minimal intervention dentistry: Part 5. Atraumatic restorative treatment (ART)-a minimum intervention and minimally invasive approach for the management of dental caries. Br Dent J 2013;214:11–18.
22. Jingarwar MM, Bajwa NK, Pathak A. Minimal intervention dentistry-a new frontier in clinical dentistry. J Clin Diagn Resch 2014;8:4–8.
23. Keilbassa AM, Gernhardt CR. Closing the gap between oral hygiene and minimally invasive dentistry : A review on the resin infiltration technique of incipient (proximal) enamel lesions. Quint Int 209;40:663–81.
24. Kim JH, Lee MA, Han GJ, Cho BH. Plasma in dentistry: a review of basic concepts and applications in dentistry. Acta Odontol Scand 2014;72:1–12.
25. Lasfargues JJ, Bonte E, Guerrieri A, Fezzani L. Minimal intervention dentistry: Part 6. Caries inhibition by resin infiltration. Br Dent J 2013;214:53–59.
26. Li X, Wang J, Joiner A, Chang J. The remineralization of enamel:a review of the literature. J Dent 2014;42:S12–20.
27. Markowitz K, Carey K. Assessing the appearance and fluorescene of resin-infiltrated white spot lesions with caries detection devices Oper Dent 2018;43:10–18.
28. McComb D. Systematic review of conservative operative caries management strategies. J Dent Educ 2001;65:1154–58.
29. Mickenautsch S, Grossman E. Atraumatic restorative treatment (ART): factors affecting success. J Appl Oral Sci 2006;14:34–39.
30. Mickenautsch S, Yengopal V, Banerjee A. Atraumatic restorative treatment versus amalgam restoration longevity: a systematic review. Clin Oral Invest 2010;14: 233–37.
31. Mittal R, Relhan N, Tangri T. Remineralizing agents: a comprehensive review. Int J Clin Prev Dent 2017;13:1–4.
32. Moune GJ, Ngo H. Minimal intervention: A new concept for operative dentistry. Quint Int 2000;31:527–31.
33. Noha A, Wassefy EL. Remineralizing effect of cold plasma and/or bioglass on demineralized enamel. Dent Mater J 2017;36:157–67.
34. Papa J, Wilson PR, Tyas MJ. Tunnel restorations:A review. J Esthet Dent 1992;4:4–8.

35. Peng JJY, Botelho MG, Matinlinna JP. Silver compounds used in dentistry for caries management: A review J Dent 2012;40:531–34.
36. Ramos-Gomez FJ, Crystal YO, Domejean S, Featherstone JDB. Minimal intervention dentistry: part 3. Paediatric dental care-prevention and management protocols using caries risk assessment for infants and young children. Br Dent J 2012;213:501–08.
37. Rao A, Malhotra N. The role of remineralizing agents in Dentistry: A review Compend 2011;32:26–33.
38. Renne WG, Mennito AS. A simplified technique for restoring interproximal root surface lesions. Oper Dent 2012;37:211–15.
39. Schwendicke F. Dorfer CE and Paris S: Incomplete caries removal: a systematic review and meta-analysis. J Dent Res 2013;92:306–14.
40. Schwendicke F. Contemporary concepts in carious tissue removal: A review J Esthet Restor Dent 2017;29:403–08.
41. Tyagi SP, Garg P, Sinha DJ, Singh UP. An update on remineralizing agents. J Interdisc Dent 2013;3:151–58.
42. Whitehouse J. Minimally invasive dentistry: Clinical application. Dent Today 2004;23:56–59.
43. Zhang Z. Mechanical benefits of conservative restoration for dental fissure caries. J Mech Behav Biomed Mater 2016;53:11–20.